Summary Table and Allowed Values for Bouhedma National Park Geospatial Database Fields

Compliant with WDPA/UNEP-WCMC Standards (Version 1.6) and WWF Talent Academy 2026 Data Dictionary (BHD26-COM-CODE)

This table defines all standardized attributes present in the GeoJSON and Shapefile layers of the Bouhedma database.

1. Official Park Zonation & Boundary (WDPA / OECM International Standard)

Requirement	Provided by	Field Name	Type	Length	Description and Allowed Values (Controlled Vocabularies)
Minimum	UNEP-WCMC	`WDPAID`	Number	N/A	Global unique identifier in the WDPA database (Bouhedma value: **13962**).
Minimum	Data provider	`PA_DEF`	Text	20	Meets IUCN and CBD protected area definitions. Allowed value: **1**.
Minimum	Data provider	`NAME`	Text	254	Official name of the protected area: *Parc national de Djebel Bou-Hedma*.
Minimum	Data provider	`DESIG`	Text	254	Name of designation: *UNESCO-MAB Biosphere Reserve & National Park*.
Complete	Data provider	`DESIG_ENG`	Text	254	Designation in English: *UNESCO-MAB Biosphere Reserve*.
Minimum	Data provider	`DESIG_TYPE`	Text	20	Designation level. Allowed value: **International**.
Complete	Data provider	`IUCN_CAT`	Text	20	IUCN management category. Allowed value: **II** (National Park).
Minimum	Data provider	`STATUS`	Text	100	Legal status. Allowed value: **Designated**.
Minimum	Data provider	`STATUS_YR`	Integer	12	Year of legal enactment: **1977** (MAB) / **1980** (Decree No. 80-1608).
Complete	Data provider	`GOV_TYPE`	Text	254	Governance type: **Federal or national ministry or agency**.
Complete	Data provider	`OWN_TYPE`	Text	254	Land ownership type: **State**.
Complete	Data provider	`MANG_AUTH`	Text	254	Management authority: *Ministry of Agriculture, Hydraulic Resources and Fisheries (DGF)*.
Minimum	GIS Calc	`REP_AREA`	Number	N/A	Reported area in square kilometres: **165.12** km² (16,512 hectares).
Minimum	Data provider	`ISO3`	Text	20	ISO 3166-1 alpha-3 code of country: **TUN** (Tunisia).
Complete	Data provider	`CONS_OBJ`	Text	254	Primary conservation objective: Protection of the last *Vachellia tortilis* savanna and reintroduction of endangered Saharan fauna.

2. Park Infrastructure, Buildings and Conservation Breeding Enclosures

Requirement	Provided by	Field Name	Type	Length	Description and Allowed Values (Controlled Vocabularies)
Minimum	Data provider	`SITE_ID`	Text	30	Unique site code for infrastructure or breeding enclosure (e.g., BHD26_CAMP01, BHD26_ENC_STCA).
Minimum	Data provider	`NAME_EN`	Text	100	Full English designation (e.g., *WWF Talent Academy Basecamp*, *French Colonial Castle Outpost*).
Complete	Data provider	`NAME_FR`	Text	100	Full French designation (e.g., *Camp de base WWF Talent Academy*).
Complete	Data provider	`TYPE`	Text	50	Structure type: **Infrastructure**, **Historical Landmark**, **Breeding Enclosure**.
Complete	Data provider	`HABITAT`	Text	10	WWF habitat controlled code (V1): **BAT** (Built structure/Camp), **ENC** (Reintroduction enclosure).
Conditional	Expert	`SPEC_CODE`	Text	10	4-letter WWF species code for enclosures: **STCA** (Ostrich), **AMLE** (Barbary sheep).
Conditional	Expert	`SPEC_SCI`	Text	80	Scientific binominal: *Struthio camelus*, *Ammotragus lervia*.
Conditional	Expert	`NB_INDIV`	Integer	10	Total individuals counted in enclosure (e.g., **9** ostriches, **3** Barbary sheep).
Conditional	Expert	`NB_MALE / NB_FEM`	Integer	10	Sex breakdown (e.g., 2 males, 1 female for Barbary sheep AMLE).
Complete	Data provider	`PURPOSE`	Text	254	Conservation role: captive breeding, chick safeguarding before open release, gene pool protection.

3. Large Herbivore & Wildlife Field Census Observations (Day 2)

Requirement	Provided by	Field Name	Type	Length	Description and Allowed Values (Controlled Vocabularies)
Minimum	Data provider	`OBS_ID`	Text	30	Unique observation code (e.g., BHD26_J2_OBS1_01).
Minimum	Data provider	`DATE`	Date	10	Observation date in ISO 8601 format (YYYY-MM-DD, e.g., **2026-09-15**).
Complete	Data provider	`TIME_UTC`	Text	30	Exact GPS observation timestamp (ISO UTC).
Minimum	Expert	`SPEC_CODE`	Text	10	4-letter WWF species code: **ORDA** (Oryx), **GADO** (Dorcas gazelle), **STCA** (Ostrich), **AMLE** (Barbary sheep).
Minimum	Expert	`SCIENTIFIC`	Text	80	Scientific binominal: *Oryx dammah*, *Gazella dorcas*, *Struthio camelus*.
Minimum	Expert	`VERNACULAR`	Text	80	Official vernacular name (Scimitar-horned Oryx, Dorcas Gazelle, North African Ostrich).
Complete	Expert	`IUCN_STAT`	Text	20	IUCN Red List status: **EW** (Extinct in the Wild / Reintroduced), **VU** (Vulnerable), **CR** (Critically Endangered).
Minimum	Expert	`INDIV_CNT`	Integer	10	Total individual head count for the observation (e.g., **1**, **2**, **10**).
Complete	Expert	`SOC_STRUCT`	Text	50	Social grouping: **Harem / Dominant Male herd**, **Solitary Male**, **Maternity Pair**.
Minimum	Data provider	`HABITAT`	Text	10	WWF habitat controlled code (V1): **SAV** (*Vachellia tortilis* savanna), **JEB** (Rocky jebel slope).
Minimum	Data provider	`BASIS_REC`	Text	10	Basis of record (V3): **VIS** (Direct visual sighting), **PHO** (Camera trap photo).
Complete	Data provider	`EXPERT`	Text	100	Scientific survey lead: **Marie Petretto** (Marwell Wildlife).

4. Specialized Sampling Stations (Bats, Camera Traps, Light Traps, Rodent Live-Traps)

Requirement	Provided by	Field Name	Type	Length	Description and Allowed Values (Controlled Vocabularies)
Minimum	Data provider	`STATION_ID`	Text	30	Standardized station code (e.g., BHD26_CT01, BHD26_BAT01, BHD26_LT01).
Minimum	Data provider	`PROTOCOL`	Text	100	Sampling methodology: **Mist Netting** (Bats), **Light Sheet Trapping** (Insects), **Live Cage Traps** (Rodents).
Complete	Data provider	`TARGET_GRP`	Text	100	Taxonomic focus: **Chiroptera**, **Carnivora**, **Insecta**, **Rodentia**.
Complete	Expert	`TARGET_SP`	Text	254	Key indicator species: RHCY, RHHI, HYHY, CAAU, VUVU, GERSP, PSOB.
Complete	Data provider	`ETHIC_RULE`	Text	254	Applied animal welfare standards: Max 5 specimens/species quota (Insects); 10–15 min net inspection (Bats); Shaded microhabitat (Rodents).
Minimum	Data provider	`EXPERT`	Text	150	Session leads: **Khaled Ferchichi** (Bats), **Dr. Haithem El-Farhati & Jamel Jrijer** (Carnivores), **Wael Ben Aba** (Insects), **Faouz Kilani** (Rodents).

5. Phytoecological Surveys, Dendrometry & Botanical Transects (Day 5)

Requirement	Provided by	Field Name	Type	Length	Description and Allowed Values (Controlled Vocabularies)
Minimum	Data provider	`OBS_ID / STAKE_ID`	Text	30	Botanical observation code (BHD26_J5_BOT_01) or reference stake (BHD26_STK_01).
Minimum	Expert	`SPEC_CODE`	Text	10	WWF botanical code: **VATO** (*Vachellia tortilis*), **STTE** (*Stipa tenacissima*), **ND** (*Not Determined / Plantae indet.* for quadrat sampling stations without single GPS-assigned species).
Minimum	Expert	`SCIENTIFIC`	Text	80	Valid taxonomic binominal: ***Vachellia tortilis*** (Umbrella Thorn Acacia), ***Stipa tenacissima*** (Esparto grass), or ***Plantae indet.*** (floristic survey detailed on field paper sheets).
Complete	Expert	`SYNONYM`	Text	100	Historical synonym noted: *Acacia raddiana* (Obsolete in Tunisian modern nomenclature).
Complete	Data provider	`METHOD`	Text	150	Quantitative technique: **Nested Quadrats (0.5m to 10m)** or **10m Fixed Dendrometric Line-Transect**.
Complet	Expert	`REMARKS`	Text	254	Recorded dendrometric metrics: Total Height (H), Crown Widths (D1, D2), Base Trunk Diameter.
Minimum	Data provider	`EXPERT`	Text	100	Phytoecology session lead: **Dr. Alaeddine Rouissi** (WWF NA).

6. Road Networks, Foot Trails and Linear Infrastructure (OSM & Field GPS)

Requirement	Provided by	Field Name	Type	Length	Description and Allowed Values (Controlled Vocabularies)
Minimum	OpenStreetMap	`OSM_ID / REF`	Text	30	OSM unique identifier or official route code (e.g., RR124 for southern asphalt regional road).
Minimum	OpenStreetMap	`HIGHWAY`	Text	30	Road classification: **secondary** (paved RR124), **service** (internal park service track), **track** (unpaved dirt road).
Complete	OpenStreetMap	`SURFACE`	Text	30	Road surface type: **asphalt** (bituminous paved), **gravel** (crushed rock track), **ground** (compacted dirt).
Minimum	Data provider	`TRAIL_ID`	Text	30	Pedestrian foot patrol trails: GPS4_TRAIL_DAY2, GPS6_TRAIL_DAY3 (zero motorized engine noise).
Minimum	OpenStreetMap	`POWER_TYPE`	Text	30	Electrical infrastructure node: **node** (power transmission pole/pylon across central savanna plain).
Complete	Data provider	`SUBSTRATE`	Text	80	Ecological function: Utility pole = primary hunting & vantage perch for raptors and shrikes.

7. Passive Acoustic Monitoring Network (PAM) & Validated Hydrology

Requirement	Provided by	Field Name	Type	Length	Description and Allowed Values (Controlled Vocabularies)
Minimum	Data provider	`SITE_ID`	Text	30	Standardized acoustic monitoring station identifier: BHD26_PAM_01 to BHD26_PAM_20.
Minimum	Data provider	`DEPLOY_ORD`	Integer	10	Sequential priority deployment order (1 to 20) ranked by descending ecological score.
Minimum	Expert	`ECOZONE`	Text	50	Stratified ecozone: **SAVANE_CENTRE**, **SAVANE_OUEST**, **SAVANE_EST**, **OUEDS_ET_PIEMONT**, **CANYON_CHARCHARA**.
Minimum	Data provider	`SUBSTRATE`	Text	80	Physical mounting structure: **Tree (Vachellia tortilis)** or **Powerline Pylon (Raptor Perch)**.
Minimum	Data provider	`LAT / LON`	Number	N/A	Geographic coordinates in WGS 84 (EPSG:4326) and projected metric coordinates in UTM Zone 32N.
Complete	SRTM DEM	`ELEV_M`	Number	N/A	Topographic elevation extracted from 30m SRTM DEM (meters above sea level).
Complete	Sentinel-2	`SAVI_DRY`	Number	N/A	Soil-Adjusted Vegetation Index in dry summer season (SAVI L=0.5, perennial canopy threshold > 0.12).
Minimum	Data provider	`DIST_WATER`	Number	N/A	Calculated continuous Euclidean distance to nearest perennial water source (meters, max 3000m).
Complete	Expert	`SCORE_ECO`	Number	N/A	Normalized composite ecological suitability score (0.000 to 1.000).
Minimum	Data provider	`WATER_NAME`	Text	80	Validated field water sources: Ain Charchara, North Cirque, WWF Camp, French Castle, Fauna troughs, Gueltas.
Complete	Data provider	`EXCL_BUF`	Number	N/A	Strict acoustic disturbance exclusion buffer: 300m setback exclusively along southern RR124 asphalt road.